



LA TROBE
UNIVERSITY

College of Science, Health and Engineering
School of Engineering and Mathematical Sciences

Object Oriented Programming Fundamentals

CSE100F

LMS Subject link:

<https://lms.latrobe.edu.au/course/view.php?id=85321>

Undergraduate

SUBJECT LEARNING GUIDE

Semester 2, 2020

Bendigo

Subject Coordinator: Dr Richard Skarbez

Instance Coordinator: Ms Rudri Kalaria

SUBJECT DETAILS

Subject Code:	CSE100F
Subject Title:	Object Oriented Programming Fundamentals
Teaching Period:	Semester 2 2020
Location(s):	Bendigo
Credit Points:	15
Delivery Mode:	Blended
AQF Level:	5
Subject relationships:	Core subject

ENROLMENT REQUIREMENTS

Prerequisites:	CSE1PE or admitted into Bachelor of Computer Science course (SBCS)
Co-requisites:	none
Assumed Skills & Knowledge:	none
Special Study Requirements:	none

Required resources: none

Additional subject requirements: none

Class materials: none

STAFF CONTACTS

Subject Coordinator:	Dr Richard Skarbez
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Tel:	03 9479 1280
Location:	238A Beth Gleeson
Consultation time/s:	By appointment
Instance Coordinator:	Ms Rudri Kalaria
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ABOUT THIS SUBJECT

Students are introduced to computers, object-oriented concepts and programming using Java. Students also gain a working knowledge of the Unix operating system. Topics covered include classes and objects, primitive data types, flow of control, methods, basic input/output and arrays. Software engineering principles are introduced, including coding standards, class design and testing strategies.

SUBJECT INTENDED LEARNING OUTCOMES (SILOs)

Upon successful completion of this subject, you should be able to:

- 1 Use an operating system and development environment to code, debug and execute Java programs that solve a given problem.

Learning Activity:

Apply Object-Oriented principles to writing programs. Evaluating errors and the corrective measures to resolve those errors.

- 2 Analyse a problem and construct a logical solution suitable for implementing as a computer program.

Learning Activity:

Students develop solutions in assignments, the real time test and the exam

- 3 Develop a basic understanding of the object-oriented concepts of objects and classes.

Learning Activity:

Initial delivery in lectures, guided practice in lecture / workshops then practical implementation in labs and assignments

- 4 Find and use existing Java components provided in code libraries.

Learning Activity:

Guided and self-discovery, initially in labs which is then expanded in assignments

- 5 Write Java code that uses control structures, classes and arrays.

Learning Activity:

Concepts presented in lectures are then developed and further exercised in labs, assignments, and lecture / workshops. Students are given help to develop the appropriate use of these structures.

- 6 Design and execute test plans.

Learning Activity:

Students are assessed on testing in assignments and the exam

GRADUATE CAPABILITIES (GC)

GC	GC Point
Literacies and Communication Skills	Not evaluated in this subject
Inquiry and Analytical Skills	Cornerstone
Personal and Professional Skills	Cornerstone
Discipline specific Knowledge and Skills	Cornerstone

ESSENTIALS

The **Essentials** are three vital areas of learning that all La Trobe undergraduate students will experience throughout their course:

1. **Global Citizenship**
2. **Innovation and Entrepreneurship**, and
3. **Sustainability Thinking**.

Information about **La Trobe's Essentials** can be found at <http://www.latrobe.edu.au/essentials/what-the-essentials-mean-for-you>.

- These areas will be introduced and evaluated in other subjects.

LEARNING ACTIVITY SCHEDULE

Week	Date	Learning Activity	Contact hours
0		Pre-reading lecture 0 (NOTE: This is examinable)	0
1	20 July	Lectures 1 & 2, Lecture/Workshop 1, Lab 1	5
2	27 July	Lectures 3 & 4, Lecture/Workshop 2, Lab 2	5
3	3 Aug	Lectures 5 & 6, Lecture/Workshop 3, Lab 3	5
4	10 Aug	Lectures 7 & 8, Lecture/Workshop 4, Lab 4	5
5	17 Aug	Lectures 9 & 10, Lecture/Workshop 5, Lab 5	5
6	24 Aug	Lectures 11 & 12, Lecture/Workshop 6, Lab 6	5
7	7 Sep	Lectures 13 & 14, Lecture/Workshop 7, Lab 7	5
8	14 Sep	Lectures 15 & 16, Lecture/Workshop 8, Lab 8	5
9	21 Sep	Lectures 17 & 18, Lecture/Workshop 9, Lab 9	5
10	28 Sep	Lectures 19 & 20, Lecture/Workshop 10, Lab 10	5
11	5 Oct	Lectures 21 & 22, Lecture/Workshop 11, Lab 11	5
12	12 Oct	Lectures 23 & 24, Lecture/Workshop 12, Lab 12	5
13	19 Oct	Real time programming test	2
		Total	62

*Timetable details are available at <http://www.latrobe.edu.au/students/timetables>. Students should also regularly check this subject's LMS site for updated information.

SELF-DIRECTED STUDY

In addition to the learning activities specified, you are expected to allocate approximately **7 hours per week of self-directed study in this subject**. This time includes subject preparation, completing learning activities, readings, assessment preparation and other related work to enhance your participation in this subject and your learning at La Trobe.

LEARNING RESOURCES & TECHNOLOGIES

Resource Type	Title	Author and Year	Publisher
Recommended			
Book	Absolute Java	Walter SAVITCH and Kenrick MOCK, 6 th Edition, 2016	Pearson
Book	Starting out with Java	Tony Gaddis, 6 th Edition, 2016	Pearson
Book	Harley Hahn's Guide to Unix and Linux	Hahn, 2008	McGraw-Hill
Book/online	Vi IMproved - VIM	Steve Oualline, 2001	New Riders, http://truth.sk/vim/vimbook-OPL.pdf

ASSESSMENT AND FEEDBACK SUMMARY

	Date due	%	Assessment Activity	Feedback method	ILOs assessed
1.	Week 6	10	Programming assignment, Part A	face-to-face marking with student in lab	1, 2, 6
2.	Week 9	15	Programming assignment, Part B	face-to-face marking with student in lab	1, 2, 4, 6
3.	Week 12	15	Programming assignment, Part C	face-to-face marking with student in lab	1, 2, 3, 6
4.	Week 13	20	1.5 hour online programming test (You will be given an 8 hour window to complete the test, with a 3 hour time limit once you begin the test.)	Marks made available on LMS	1, 2, 3, 4, 5
5.	Week 8	20	1-hour online (LMS) examination (You will be given a 8 hour window to complete the exam.)		2, 3, 5, 6
6.	Exam Period	20	1-hour online (LMS) examination (You will be given a 4 hour window to complete the exam.)		2, 3, 5, 6

* Penalties apply if you submit an assessment task after the due date. Refer to the **Assessment (Late Submission of Tasks) Policy** at <https://policies.latrobe.edu.au/document/view.php?id=148&version=1> for further information.

**** Students should keep a copy of all assignments submitted for assessment.**

Additional information about assessment & feedback

All programming assignments and the Real Time programming test are to be submitted on latcs8, using the submit command. This is explained in every assignment, in detail. If you are unsure how to use the submit command, please consult with your lecturer or your demonstrators.

All programming assignments and the Real Time test **MUST** compile and run on the Unix latcs8 environment. Code that compiles and runs on other IDE's or operating systems, but not on latcs8, will not be marked, regardless of the correctness of the code. Please make sure that your code compiles and runs on latcs8. You are, after all, required to make your programs run on the client's environment, not expect the client to adapt to you.

All programming assignments can be solved with the material presented up to the point where they are due in. Most assignments have direct links to a small number of labs that form the preliminary work for the assignment.

Whilst we encourage students to push ahead, above all, understanding the code that you have submitted is the critical requirement. If you understand your code, you will be able to reproduce the code in the appropriate form and place in later assignments, the exam and the real time programming test.

For example, the exam will use the same problem domain in section B, the long programming question, from Assignment Part C and the Progress Check Test.

Not being able to explain code that you have submitted will result in your assignment losing marks, up to and including being awarded a mark of 0. This is regardless of the correctness of the execution of the submitted code.

Students are encouraged to submit code that does not fully compile. At the face-to-face marking, the marking staff, may, be able to identify some small errors, correct them and then get the program compiling and give the possibility of gaining most of the marks. This is at the discretion of the marking staff and the time available. Programs that generate more errors after the first errors are fixed will more than likely end up being awarded 0.

Your assignments will be checked for plagiarism, that is, a very high similarity between 2 or more students programs, by the University code plagiarism checkers. **It is your responsibility to have your assignment submitted by the assessment due date. Further action will be taken if the plagiarism checker report indicates a prima facie case of plagiarism.**

HURDLE REQUIREMENTS

There are a variety of assessment hurdles or conditions for passing a subject other than the overall mark. **Hurdle Requirements** may be built into the assessment of the learning outcomes of your subject.

There are no hurdle requirements for this subject this semester.

SPECIAL CONSIDERATION

If you are not able to complete an assessment task on the scheduled date due to exceptional circumstances beyond your control, you may be eligible for **Special Consideration**.

If the assessment task is worth 15% or more of the total assessment for this subject and you have experienced serious short term, adverse and unforeseen circumstances that substantially affect your ability to complete an assessment task to the best of your potential, you may be eligible to apply for **Special Consideration**. To do this or find more information, go to <http://www.latrobe.edu.au/special-consideration>

For Special Consideration for assessment tasks worth less than 15% of your total mark, contact your Subject Coordinator.

For Information about the Special Consideration policy, go to **Special Consideration Policy** at <https://policies.latrobe.edu.au/document/view.php?id=205>

REQUEST AN EXTENSION OF TIME TO SUBMIT AN ASSIGNMENT TASK

Penalties apply if you submit an assessment after the due date.

If the assessment task is worth 15% or more of the total assessment for this subject and you are not able to submit it by the due date, you will need to **request an extension of time to submit**. To do this or find more information go to <http://www.latrobe.edu.au/students/admin/forms/request-an-extension>

For extensions of time for assessment tasks worth less than 15% of your total mark, contact your subject coordinator. **Poor time management is not an acceptable reason for an extension.**

For information about late submission policy, go to **Assessment (Late Submission of Assessment Tasks) Policy** at <https://policies.latrobe.edu.au/document/view.php?id=148&version=1>

ACADEMIC INTEGRITY

“Academic integrity means being honest in academic work and taking responsibility for learning the conventions of scholarship. Academic integrity education is integral to the learning experience at La Trobe University. The University requires its academic staff and students to observe the highest ethical standards in all aspects of academic work, and it demonstrates its commitment to these values by awarding due credit for honestly conducted scholarly work, and by penalising academic misconduct and all forms of cheating”. (La Trobe University Academic Integrity – Procedures and Responsibilities, 2015).

The penalty for submitting an assignment as your own but is the work of a third-party, may be as severe as “*exclusion from the University without readmission*”. Refer to the **Academic Integrity - Schedule of Responses and Penalties for Academic Misconduct** at

https://www.latrobe.edu.au/__data/assets/pdf_file/0006/847923/Academic-Integrity-Schedule-of-Responses.pdf

You should familiarise yourself with the **Academic Integrity** website at <http://www.latrobe.edu.au/students/academic-integrity>, and complete the **Academic Integrity Module (AIM)** in your LMS.

If you have any questions regarding **Academic Integrity**, your Subject or Course Coordinator will be able to assist.

STUDENT FEEDBACK ON SUBJECT SURVEYS

Student feedback contributes to enhancing the student experience and to improving the quality of learning and teaching at the University. The University piloted two new survey instruments in 2019 which will be rolled out for all 2020 subjects. The new student feedback instruments include:

1. **Early Feedback on Learning** – You will receive this survey early in your course. The purpose of this survey is for you to highlight what works well or does not while there is still time to make changes in the semester. Teaching staff will review your feedback and introduce changes where appropriate to further improve this subject and enhance your learning.
2. **Subject Quality Survey** - This survey allows you to provide feedback prior to the end of the teaching period about this subject and the teaching of this subject. It combines the previously separate Student Feedback on Subjects (SFS) and Student Feedback on Teaching (SFT) surveys. The Subject Quality Survey (SQS) is one of the quality assurance processes that occur across the University. While the survey is anonymous, we do expect that you use respectful language and provide constructive feedback. Your views are taken seriously and assist us to improve the learning experience for future students. Feedback from previous students who completed this subject, as well as the action taken by the subject coordinator in response to the feedback is available in the LMS site for this subject.

The new survey instruments are shorter than previous surveys and have been designed in conjunction with both staff and students.

CHARTER OF STUDENT RIGHTS AND RESPONSIBILITIES

As a La Trobe coursework student, you have a number of rights and responsibilities. La Trobe statutes and regulations, as well as policies and procedures govern your rights and responsibilities. La Trobe University aspires to provide an outstanding student experience conducive to scholarly activity and supportive of your learning goals.

The **Charter of Student Rights and Responsibilities** is a brief summary of your rights and responsibilities as a student at La Trobe. The Charter was developed in consultation with La Trobe students. A copy of the **Student Rights and Responsibilities** can be found at <https://policies.latrobe.edu.au/document/view.php?id=225>.

STUDENT COMPLAINTS

The **Student Complaints Management System (SCM)** is a formal mechanism for students to lodge concerns about any aspect of their University experience. A student complaint may be about any aspect of University life, such as an administrative matter, a problem with a subject or course, or the conduct of a student or staff member.

For more information about the **Student Complaint process** and how to lodge a complaint, visit the **Complaints and Feedback** page at: <https://www.latrobe.edu.au/students/support/complaints>.

GETTING HELP AND SUPPORT AT UNI

A range of services at La Trobe can assist you with your studies, health and wellbeing.

For the full range of services and facilities available for you, go to **Student Help and Support** at <http://www.latrobe.edu.au/students/support>.

LIBRARY

Your **Library** offers many ways to support your study. To discover more about how the Library can help, visit your **Library website** and explore the following essential resources:

- **Achieve@Uni**
<https://www.latrobe.edu.au/students/study-resources/learning/achieve> - Learn skills to help complete your assessments
- **Assessment Planner**
<http://www.latrobe.edu.au/library/assessment-thesis-support/plan-my-assessment> – Create a personal assessment schedule
- **Expert Help Guides**
<https://latrobe.libguides.com/> - Find expert discipline and specialist help for your research
- **Referencing at La Trobe University**
<http://latrobe.libguides.com/referencing/> - Understand how and why to reference
- **Academic Referencing Tool**
www.lib.latrobe.edu.au/referencing-tool - Get detailed examples of referencing styles
- **Chat to library staff**
www.latrobe.edu.au/library - Access the text chat service from the Library website
- **Contact us**
www.latrobe.edu.au/library/about/contact - Call or email your Library

ASK LA TROBE

ASK La Trobe is a service open to all La Trobe students who need guidance with administrative matters during their time at La Trobe.

ASK La Trobe can provide you with help relating to enrolments and timetabling, fees and financial support, exams, results and graduations & student life and support services.

Go to **ASK La Trobe – Current Students** at <http://latrobe-current.custhelp.com/app/home> and start by looking at our **FAQs**, or visit us at your campus **ASK La Trobe Help Zone**.

If you can't find what you're looking for:

- **Call** 1300 La Trobe (1300 528 762)
- **Ask a question** online at <http://latrobe-current.custhelp.com/app/ask>
- **Visit the ASK La Trobe Help Zone.** Join the queue for the Bundoora, Bendigo or city campuses using the **QLess: Virtual Queuing system**.

LEARNING AND SUPPORT PROGRAMS

There are many resources, including services, to help you with your academic studies at La Trobe. Go to the **Learning and Support Programs** page at <http://www.latrobe.edu.au/students/study-resources/learning> for further information.

HEALTH AND WELLBEING

Your health and wellbeing are an important part of being a successful student. Studying can be stressful and many factors can affect your mental and physical health, in and outside of the University.

Check out the services available to you at La Trobe to support your health and wellbeing at www.latrobe.edu.au/wellbeing.